

Localize, don't classify: detection survives the study shift that collapses classification

Schistosomiasis (mar2020→nov2021, brightfield) + Chula-ParasiteEgg-11 (11 species) · summary for review · 2026-06-14

Summary

Automated microscopy diagnosis is usually framed as **image classification** (is this sample positive?). On schistosomiasis brightfield microscopy across two field studies (train mar2020, zero-shot test nov2021), an ImageNet-pretrained classifier learns the source study well (within-study AUC ≈ 0.85) but **collapses to chance on the unseen study (AUC ≈ 0.52 , 95% CI through 0.50)**. Re-framing the task as **object detection** (localize the egg, aggregate per patient) is robust to the same shift: a YOLOv8 detector on the *same* patients holds at AUC 0.914 [0.859, 0.961] — **reproducing the established cross-study result of de Leon Derby et al. (2025)** (their BF M&E sensitivity 76%; ours 73.8%), not beating it. Saliency shows the mechanism: the classifier attends to global background / illumination, not the egg. On a second dataset (Chula, 11 species), a class-agnostic egg detector trained on 8 species generalizes to 3 unseen species (mAP50 0.993 vs 0.992). **The contribution is the contrast + mechanism + cross-species generalization — localizing the invariant target is robust to shift (site shift and novel-species shift) that breaks global classification.**

Result 1 — classification collapses cross-study; detection does not (same direction, same patients)

Model	within-study (mar val)	cross-study (nov, zero-shot)
MobileNetV2	0.854	0.517 [0.432, 0.600]
EfficientNet-B0	0.857	0.504 [0.426, 0.590]
YOLOv8s detector	—	0.914 [0.859, 0.961]

380 nov2021 patients (61 pos / 319 neg). Detector M&E sensitivity 0.738 @ 96.5% spec (de Leon Derby BF: 0.76); TI&S 0.557 @ 99.5% spec. Classifier and detector CIs do not overlap. 2000-sample bootstrap CIs.

Result 2 — cross-species generalization (Chula-ParasiteEgg-11)

	in-distribution (8 seen)	out-of-distribution (3 unseen)
mAP@50	0.992	0.993
mAP@[.5:.95]	0.943	0.896

Figures

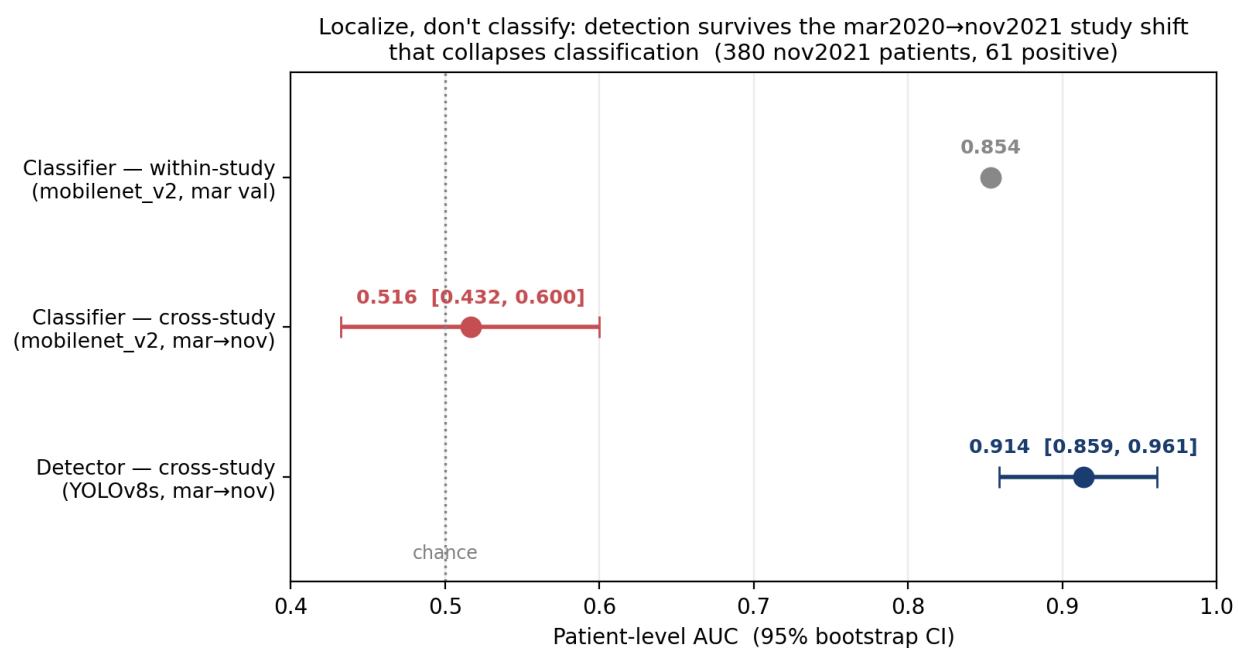


Fig 1. Same patients, same shift: the classifier learns mar2020 (0.85) but collapses to chance on nov2021 (0.52); the detector holds (0.914).

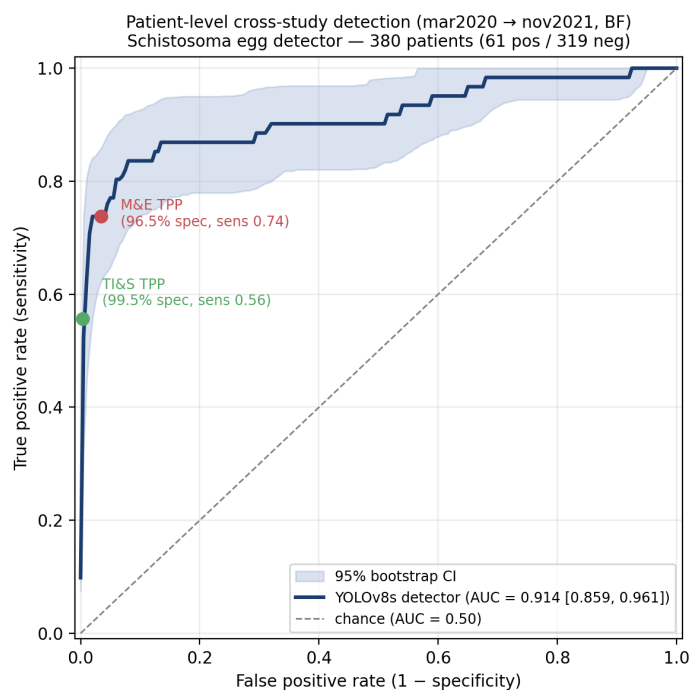


Fig 2. Patient-level ROC of the detector (mar→nov, bootstrap CI band, M&E/TI&S; operating points).

Mechanism: the mar→nov classifier attends to global background / illumination, not the egg
(an egg is ~8 px in this 224-px view) — the invariant cue the detector localizes

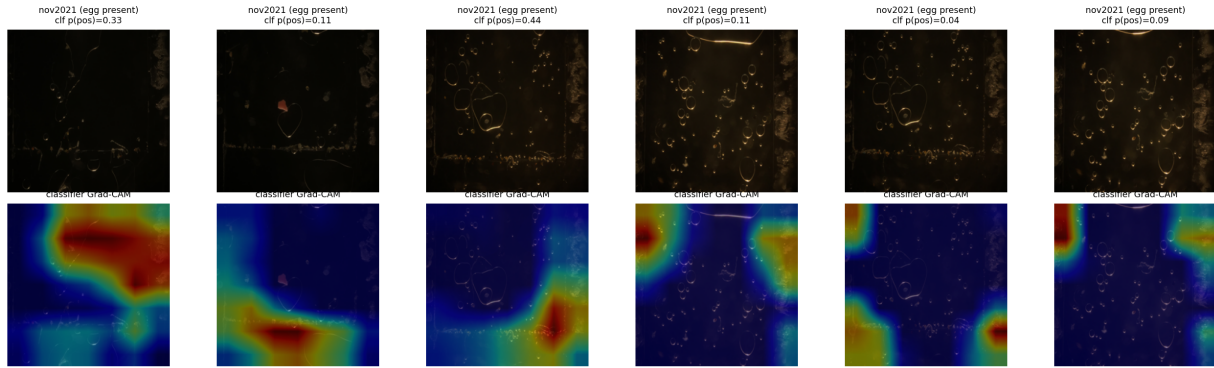


Fig 3. Mechanism — Grad-CAM of the cross-study classifier on egg-present nov2021 images: attention on background/illumination, not the egg.

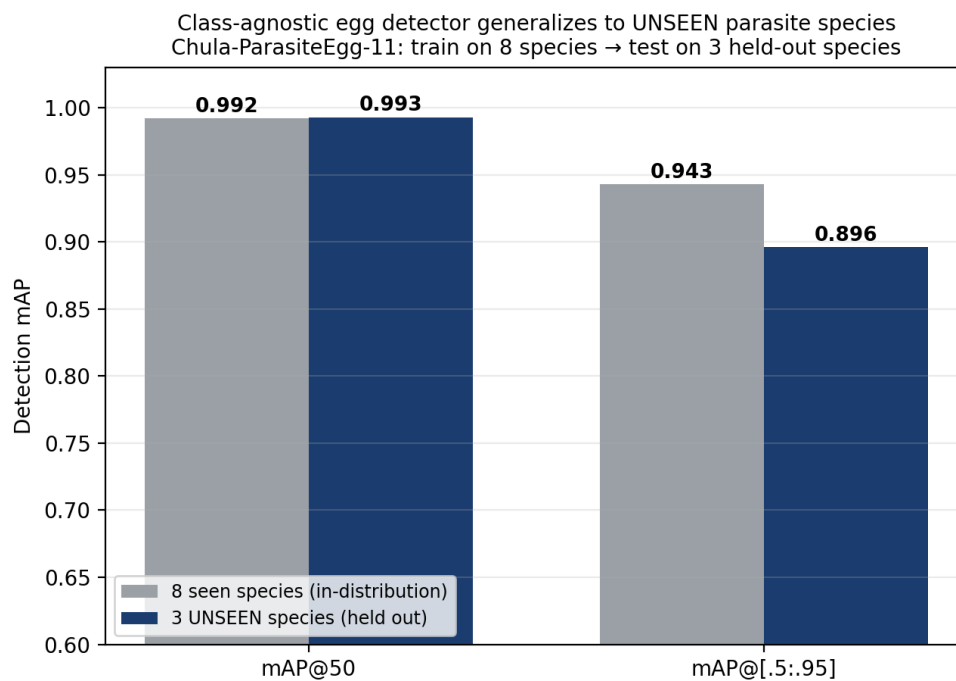


Fig 4. Second disease: a single-class detector trained on 8 species localizes 3 unseen species (mAP50 0.993 vs 0.992 in-distribution).

Honest limitations

- The high-level principle (classifiers exploit shortcuts and fail out-of-distribution) is known in ML; the contribution is the rigorous clinical demonstration + mechanism + cross-disease generalization, not a new phenomenon.
- The detector result reproduces de Leon Derby et al. (2025) (same March→Nov split); it is not claimed to beat prior work.
- Chula detection is near-saturated/easy (mAP50 0.99 in-distribution); the honest signal is the mAP50-95 drop (0.943→0.896). One random fold of held-out species.
- Headline contrast is one direction (mar→nov); detector within-study (mar→mar) not yet measured. Mechanism is qualitative.